

Modeling the Relationship Between Neutron Counting Rates and Sunspot Numbers Using the Hysteresis Effect

F. Inceoglu · M.F. Knudsen · C. Karoff · J. Olsen

Received: 7 May 2013 / Accepted: 10 August 2013 / Published online: 28 September 2013
© Springer Science+Business Media Dordrecht 2013

Abstract Several studies show that temporal variations in the Galactic cosmic ray (GCR) intensity display a distinct 11-year periodicity due to solar modulation of the galactic cosmic rays in the heliosphere. The 11-year periodicity of GCRs is inversely proportional to, but out of phase with, the 11-year solar cycle, implying that there is a time lag between actual solar cycle and the GCR intensity, which is known as the hysteresis effect. In this study, we use the hysteresis effect to model the relationship between neutron counting rates (NCRs), an indicator of the GCR intensity, and sunspot numbers (SSNs) over the period that covers the last four solar cycles (20, 21, 22, and 23). Both linear and ellipse models were applied to SSNs during odd and even cycles in order to calculate temporal variations of NCRs. We find that ellipse modeling provides higher correlation coefficients for odd cycles compared to linear models, *e.g.* 0.97, 0.97, 0.92, and 0.97 compared to 0.69, 0.72, 0.53, and 0.68 for data from McMurdo, Swarthmore, South Pole, and Thule neutron monitors, respectively, during solar cycle 21 with overall improvement of 31 % for odd cycles. When combined to a continuous model, the better correlation observed for the odd cycles increases the overall correlation between observed and modeled NCRs. The new empirical model therefore provides a better representation of the relationship between NCRs and SSNs. A major goal of the ongoing research is to use the new non-linear empirical model to reconstruct SSNs

F. Inceoglu (✉) · C. Karoff
Stellar Astrophysics Centre, Department of Physics and Astronomy, Aarhus University, Ny Munkegade
120, 8000 Aarhus C, Denmark
e-mail: fadil@phys.au.dk

C. Karoff
e-mail: karoff@phys.au.dk

M.F. Knudsen
Department of Geoscience, Aarhus University, Høegh-Guldborgs Gade 2, 8000 Aarhus C, Denmark
e-mail: mfk@geo.au.dk

J. Olsen
AMS, ^{14}C Dating Centre, Department of Physics, Aarhus University, Ny Munkegade 120, 8000
Aarhus C, Denmark
e-mail: jesper.olsen@phys.au.dk

on annual time scales prior to 1610, where we do not have observational records of SSNs, based on changes in NCRs reconstructed from ^{10}Be in ice cores.

Keywords Solar-terrestrial relations · Sun: activity · Sun: dynamo · Sunspots

1. Introduction

Galactic cosmic rays (GCRs) are charged particles with energy ranges from a few MeV to approximately 10^{20} eV, impinging on the Earth from all directions. The majority of cosmic ray particles are atomic nuclei, but electrons, positrons, and other sub-atomic particles are also included. At the top of the Earth's atmosphere, the composition of GCRs includes protons (87 %), alpha particles (12 %), and heavier nuclei (1 %) (Dunai, 2010). Upon entering the atmosphere, these particles interact with atmospheric atoms and produce secondary particles, primarily neutrons, via spallation and subsequent cascade processes. In order to observe the GCR intensity, neutron monitors were established at different latitudes on the Earth since the first half of the 20th century.

GCRs in the heliosphere are modulated by changes in the solar magnetic activity, and the number of sunspots on the surface of the Sun represents a widely used indicator of these activity variations. Sunspot numbers (SSNs) have been used to study past solar activity (Hoyt and Schatten, 1993; Lean, 2000; Solanki *et al.*, 2004) and to predict the amplitudes of the forthcoming solar cycles (Owens, Usoskin, and Lockwood, 2012), particularly the 24th solar cycle, which is the subject of much debate due to the extended minimum phase preceding this cycle (see *i.e.* Hathaway (2010) for a recent review).

Information on solar variations prior to 1610 relies on past production rates of cosmogenic nuclides, such as ^{10}Be and ^{14}C (Bard and Frank, 2006). Cosmogenic nuclides are produced in the atmosphere by interactions of GCR particles from space with atmospheric elements, such as N and O. Their production rates are inversely correlated with solar magnetic activity and the geomagnetic field intensity due to the non-linear shielding effect of the solar magnetic field and the geomagnetic dipole field (Heikkilä *et al.*, 2008).

Therefore, a more detailed understanding of the relationship between SSNs and neutron counting rates (NCRs), as indicator of GCRs, is needed since the time resolution of past cosmogenic nuclide production-rate reconstructions based on ice cores (^{10}Be) and tree rings (^{14}C) becomes increasingly higher (Baroni *et al.*, 2011; Owens, Usoskin, and Lockwood, 2012). Using such high-resolution records, it is possible to reconstruct GCR intensity further back in time prior to the instrumental measurements of NCRs. Information on the short- and long-term behavior of the GCR intensity obtained from cosmogenic nuclide production-rate reconstructions will make it possible to reconstruct SSNs beyond 1610 AD and hence improve our understanding of the long- and short-term variations in solar activity cycles on millennial time scales.

In this study, we investigate the time lag between monthly SSNs and NCRs, as an indicator of GCR intensity, over a period that covers the last four solar cycles and we introduce a new model that links NCRs and SSNs and makes it possible to calculate temporal changes of NCRs based on the hysteresis effect. The layout of the paper is as follows. In Section 2, we describe the NCR data that we use. In Section 3, we study the hysteresis phenomenon. We then model the relationship between NCRs and SSNs in Section 4 and discussion and conclusions are found in Section 5.

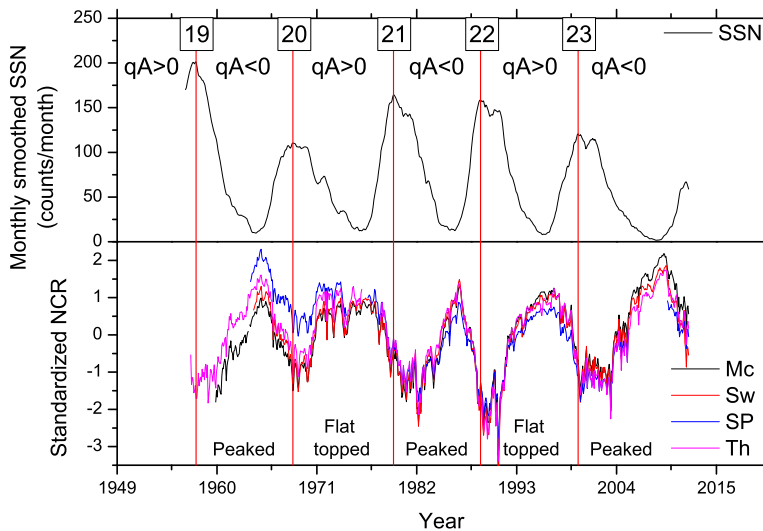


Figure 1 SSNs (top) and NCRs recorded by neutron monitors (bottom). Mc, Sw, SP, and Th denote McMurdo, Swarthmore, South Pole, and Thule neutron monitors, respectively.

2. Data Analysis

In order to study the long-term GCR modulation in the heliosphere over a time period covering solar cycles 20, 21, 22, and 23, we used pressure-corrected neutron counting rates (NCRs) obtained from four different neutron monitors (NMs). Three of these monitors (South Pole in Antarctica, McMurdo in Antarctica, and Thule in Greenland) are located close to the poles and have cut-off rigidities of 0.1–0.3 GV and thus record GCRs at essentially all energies, whereas one (Swarthmore) is located in Pennsylvania (39.9°N 75.4°E) and therefore has a cut-off rigidity of 2.4 GV. Since these stations are located around the globe at various geographical locations and altitudes, differences exist among the counting rates recorded at various neutron monitors (Usoskin *et al.*, 1998, 1999, 2001), but these differences do not affect the results presented in this paper. Additionally, it should be emphasized that there is a long-term trend in the counting rates of the South Pole NM, which may be due to improper normalization of its data and difficulties relating to maintenance of the NM (Ahluwalia and Ygbuhay, 2012). SSN data throughout the last four solar cycles were obtained from the Solar Influences Data Center.¹

The pressure-corrected neutron counting rates from four different neutron monitors were standardized using their individual mean and standard deviation values. The standardization was used for plotting only. Temporal changes in neutron counting rates and monthly smoothed sunspot numbers are shown in Figure 1.

Although a strong correlation is observed between SSNs and NCRs it is also clear that the relationship between NCRs and SSNs varies from cycle to cycle. Especially, it is clear that NCRs show flat-topped maxima during transitions between declining phases of even cycles and inclining phases of odd solar cycles, whereas peaked-topped cycles are seen during transitions between declining phases of odd solar cycles and inclining phases of even solar

¹<http://sidc.oma.be/sunspot-data/>.

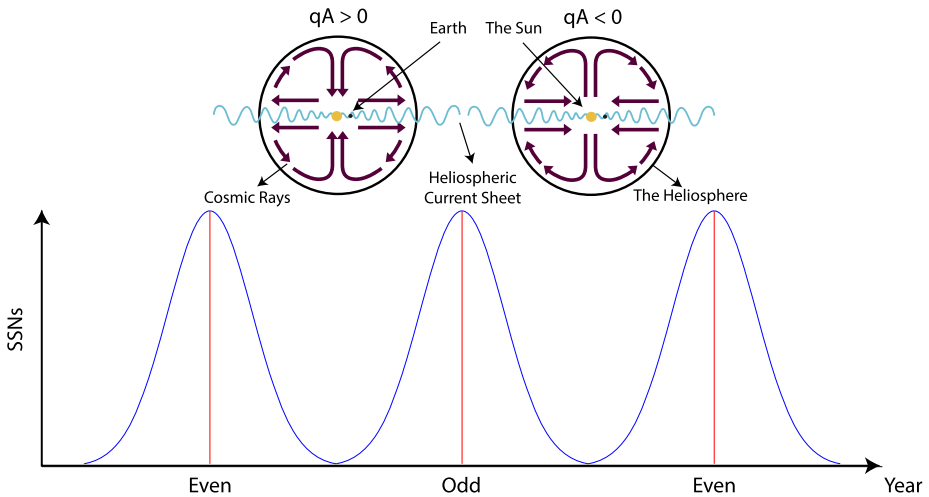


Figure 2 Illustration of trajectories of GCRs in the heliosphere under different solar polarity conditions (modified from Jokipii, 2011).

cycles (Webber and Lockwood, 1988). This phenomenon is related to differences in the way the GCRs are modulated in the heliosphere during odd and even cycles, respectively, but no definitive explanation exists (Van Allen, 2000).

It is well known that the 11-year solar cycle is in fact a 22-year cycle, when the polarity of the large-scale solar magnetic field is taken into account (Hale *et al.*, 1919). The varying relationships between NCRs and SSNs during odd and even cycles relate to the polarity of the large-scale solar magnetic field. The difference can be explained by including a drift term in the model due to Parker (1965) of the modulation of GCRs through the heliosphere as suggested by Jokipii, Levy, and Hubbard (1977) and Van Allen (2000). In this expanded model, the positively charged GCR particles ($q > 0$) drift inward from the heliographic poles to the equator and outward along the current sheet when the polarity of the large-scale solar magnetic field is positive ($qA > 0$), meaning that the direction of the polarity is aligned with the solar rotation axis (Figure 2). When the polarity of the large-scale magnetic field is negative ($qA < 0$), the GCR particles experience an inward drift along the current sheet (Mavromichalaki *et al.*, 1996; Mavromichalaki, Belehaki, and Rafios, 1998) and therefore mainly enter the heliosphere through the equatorial regions. However, where the GCR particles experience an outward drift during the declining phase of even cycles and the onset of odd cycles ($qA > 0$), they will experience an inward drift during the declining phase of odd cycles and onset of even cycles ($qA < 0$) (Figure 2). This will lead to a slow recovery, which means longer time delays between changes in SSNs and GCRs (Van Allen, 2000) after maxima in odd cycles and a fast recovery after maxima in even cycles. In other words, the time it takes the GCRs to reach the Earth is smaller during the final phase of even cycles and the initial phase of odd cycles (Belov, 2000; Potgieter, Burger, and Ferreira, 2001; Singh, Singh, and Badruddin, 2008), resulting in a smaller delay between solar variability and cosmic ray particles reaching the Earth during even cycles. In addition, the observed peaked and flat-topped maxima in the temporal variation of NCRs are thought to be linked to these slow and fast recovery times (Van Allen, 2000; Ahluwalia and Ygbuhay, 2012). Although this agrees with what is observed in Figure 1, where we also mark the polarity state of the large-scale solar magnetic field during the different cycles, it is not the only explanation for

Figure 3 Cross-correlation coefficients between monthly NCRs and SSNs as a function of time lag.

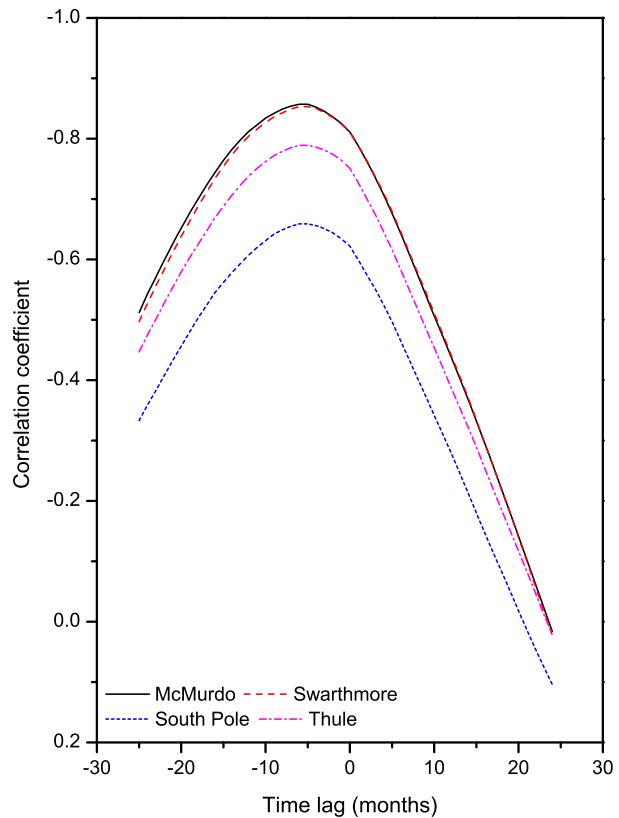


Table 1 Cross-correlation coefficients and the corresponding time lags for each of the four neutron monitors for the period from 1957 to 2012.

Stations	Time lag (months)	Correlation coefficient	
		Without time lag	With time lag
McMurdo	5	-0.81	-0.86
Swarthmore	5	-0.81	-0.85
South Pole	5	-0.62	-0.66
Thule	6	-0.75	-0.79

the varying, cycle-dependent relationships between NCRs and SSNs. Other explanations include *e.g.* the alignment between the polarity of the large-scale solar magnetic field and the galactic field (Nagashima and Morishita, 1980; Otaola, Perez-Enriquez, and Valdes-Galicia, 1985). Additionally, it was found that GCRs experience a stronger modulation during positive solar polarity state conditions, whereas GCR particle modulation becomes relatively weaker during negative polarity states of the Sun (Alanko-Huotari *et al.*, 2007).

To investigate this relationship and the long-term modulation of GCRs, cross-correlation analysis was applied to the data covering the last four solar cycles, using a time window of width $T = 50$ months centered at time t . Following the approach of Usoskin *et al.* (1998), we then calculated the cross-correlation for data shifted between $t - T/2$ and $t + T/2$. The analysis yields an average time lag of 5 months between the SSNs and the GCRs (Figure 3 and Table 1).

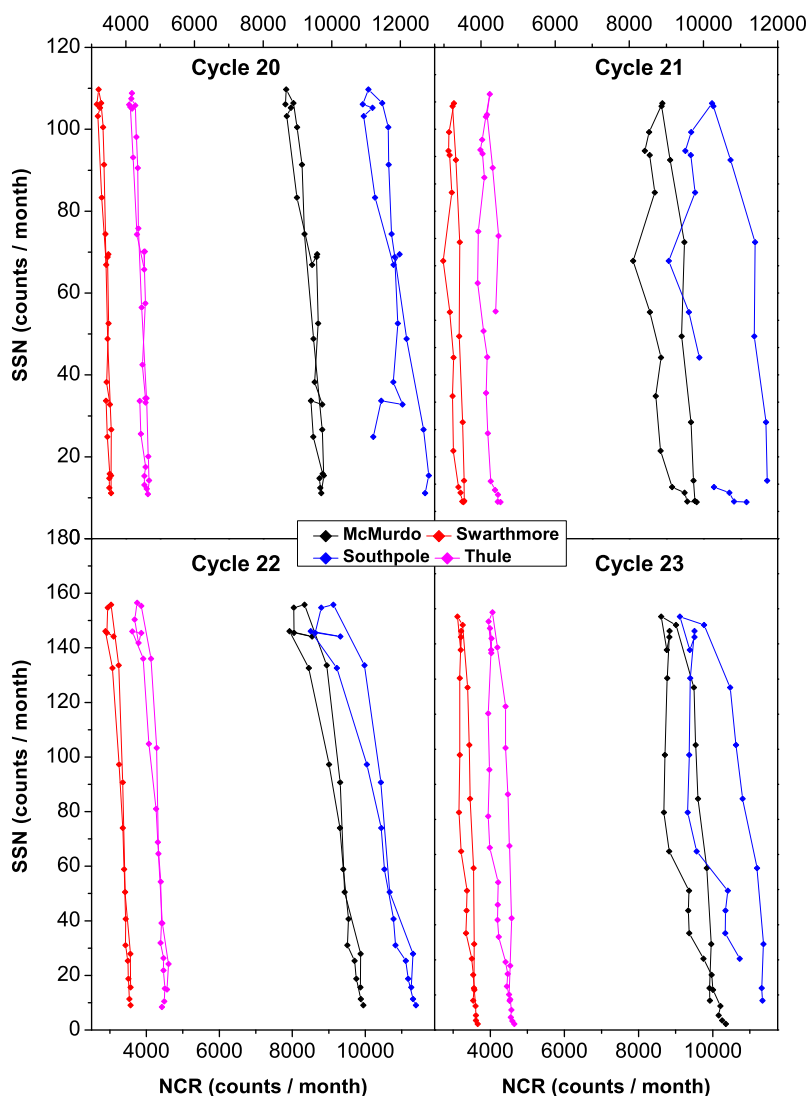


Figure 4 Hysteresis plots between NCRs as recorded by neutron monitors and SSNs for solar cycles 20–23. Data sets are averaged over a period equal to the 6-month time lag for plotting purposes.

3. Hysteresis Effect

The hysteresis effect and time lag between long-term variations in GCR intensity and solar activity have been reported before by various authors (Mavromichalaki and Petropoulos, 1984; Marmatsouri *et al.*, 1995; Gupta, Mishra, and Mishra, 2006; Mavromichalaki, Paouris, and Karalidi, 2007). The underlying mechanism that causes this time lag is explained in detail above.

In this study, we use the hysteresis effect to reconstruct NCRs based on SSNs. To obtain hysteresis curves that properly take into account the average long-term time lag, each data set derived from the different NMs was averaged over a period equal to the 5-month time

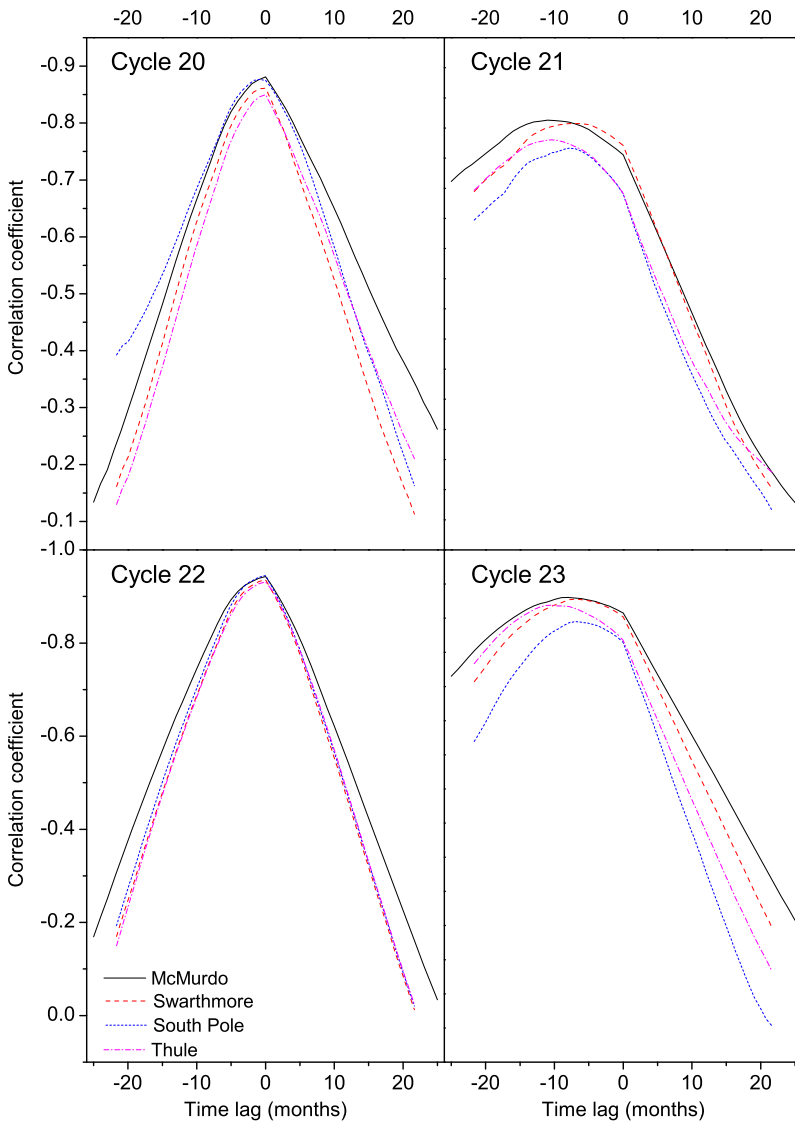


Figure 5 Cycle-wise cross-correlation coefficients between monthly NCRs and SSNs as a function of time lag.

lag for plotting purposes. The hysteresis curves show that even cycles and odd cycles have different loop shapes and different time lags (Figure 4). We thus calculated the time lag for each individual solar cycle between two SSN minima (Figure 5 and Table 2). As clearly shown in Table 2, all the neutron monitors exhibit similar behavior regarding the hysteresis effect. During even cycles, almost no time lag is observed except for the South Pole station data (1 month). On the contrary, during odd cycles, the time lag varies from 8 months to 12 months among the NM stations. The time lag found for odd cycles are in good agreement with the calculated time lag of 13–14 months for solar cycle 23 in previous works

Table 2 Cross-correlation coefficients and the corresponding time lags for each of the solar cycles 20, 21, 22, and 23 separately and for the four neutron monitors.

Stations	Time lag (months)	Correlation coefficient	
		Without time lag	With time lag
Cycle 20			
McMurdo	0	−0.88	−0.88
Swarthmore	0	−0.87	−0.87
South Pole	1	−0.80	−0.80
Thule	0	−0.85	−0.85
Cycle 21			
McMurdo	11	−0.61	−0.73
Swarthmore	9	−0.64	−0.72
South Pole	9	−0.48	−0.63
Thule	12	−0.48	−0.66
Cycle 22			
McMurdo	0	−0.94	−0.94
Swarthmore	0	−0.94	−0.94
South Pole	0	−0.95	−0.95
Thule	0	−0.93	−0.93
Cycle 23			
McMurdo	8	−0.82	−0.86
Swarthmore	8	−0.80	−0.86
South Pole	8	−0.73	−0.79
Thule	12	−0.74	−0.84

(Mavromichalaki, Paouris, and Karalidi, 2007; Paouris *et al.*, 2012) and slightly different from the results of Chowdhury, Kudela, and Dwivedi (2013) who obtained a time lag of 18 months. The reason for this difference is probably related to the fact that the latter estimate is based on NCR data from another NM with a different cut-off rigidity of 0.81 GV. The estimated time lag of Chowdhury, Kudela, and Dwivedi (2013), based on data from the Oulu NM (65.05°N 25.47°E), was computed using 27-day-averaged GCR intensity data, whereas our analyses are based on monthly averaged data.

4. Modeling the Relationship Between NCRs and SSNs

Prior to the modeling, the NCR data were linearly interpolated over the last four solar cycles to ensure continuity in the time series. This was necessary because small gaps exist in the data recorded by various neutron monitors, *e.g.* March 1978 for Swarthmore NM, July 1976 to June 1978 (24 months) for Thule NM, and November 1974 to January 1977 (27 months) and December 2005 to December 2008 (36 months) for South Pole NM. Additionally, in order to get rid of high-frequency fluctuations in the data sets, we smoothed the data in the time domain using a 7-month moving average. However, to make a comparison, we present analyses of both the smoothed and non-smoothed data series.

In order to reconstruct NCRs using SSNs, we have derived cycle-wise relationships based on the hysteresis effect. Since the link between SSNs and NCRs depends on whether the solar cycle is even or odd, this relationship also varies as a function of time. For even cycles, characterized by narrow hysteresis loops (Figure 4), linear regression was used, whereas for

Figure 6 The hysteresis curves of even cycles (20 and 22) and corresponding linear fits. Orange and green diamonds show extended (see the text) and interpolated data, respectively.

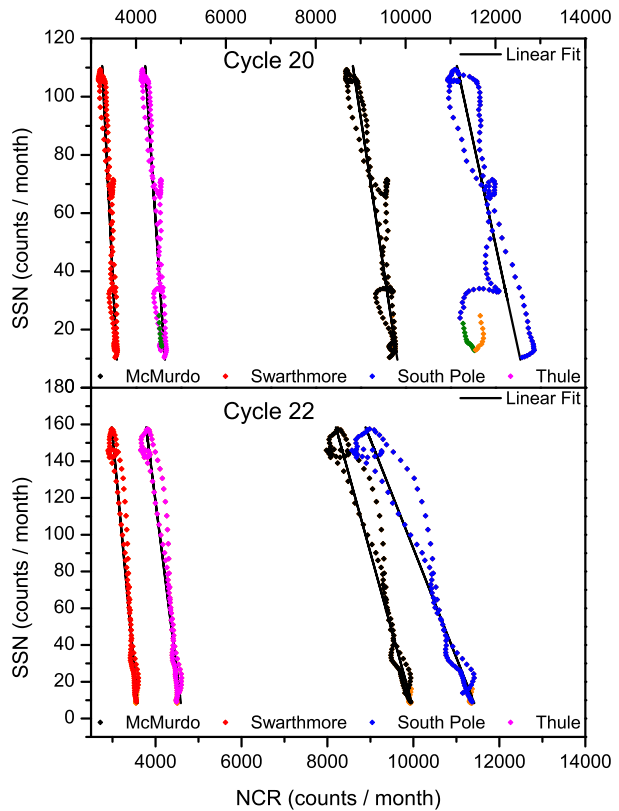


Table 3 The parameters and correlation coefficients of linear modeling of NCRs for even cycles. The first value in each row shows the result for smoothed data, whereas the second value represents result for non-smoothed data. The linear regression equation is as follows: $NCR = \alpha SSN + \beta$.

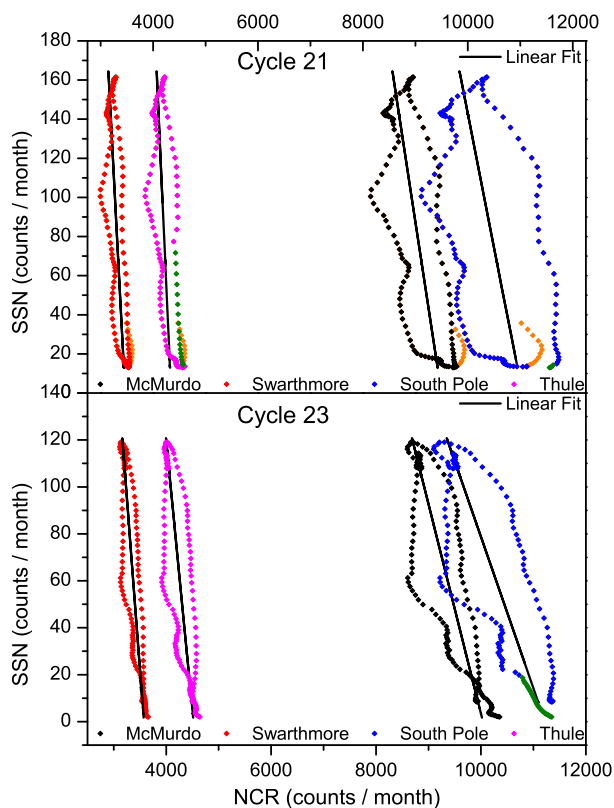
Station	α	β	Correlation coefficient
smoothed / unsmoothed			
Cycle 20			
McMurdo	9911/9912	-9.8/-9.8	0.91/0.88
Swarthmore	3604/3605	-3.2/-3.2	0.90/0.87
South Pole	12131/12137	-8.0/-8.0	0.58/0.57
Thule	4675/4675	-4.2/-4.2	0.88/0.85
Cycle 22			
McMurdo	10053/10051	-12.0/-12.0	0.97/0.94
Swarthmore	3610/3610	-4.0/-3.9	0.96/0.94
South Pole	11549/11545	-17.0/-17.0	0.97/0.94
Thule	4622/4620	-5.3/-5.3	0.96/0.93

odd cycles, characterized by wider hysteresis loops (Figure 4), both linear and ellipse fitting were used to determine the best approach to calculate the relationship between NCRs and SSNs. In order to provide a smooth transition between odd and even cycles, we used a 12-month-overlapping period by extending the decreasing phase of the solar cycles by including the first 12 months of the subsequent cycle. Following this, we calculated weighted averages

Table 4 The parameters and correlation coefficients of linear modeling of NCRs for odd cycles. The first value in each row shows the result for smoothed data, whereas the second value represents result for non-smoothed data. The linear regression equation is as follows: $NCR = \alpha SSN + \beta$.

Station	α	β	Correlation coefficient
smoothed / unsmoothed			
Cycle 21			
McMurdo	9609/9609	-6.6/-6.5	0.69/0.66
Swarthmore	3500/3500	-2.2/-2.2	0.72/0.69
South Pole	11084/11084	-8.0/-8.0	0.53/0.52
Thule	4485/4485	-2.7/-2.7	0.68/0.66
Cycle 23			
McMurdo	10039/10038	-11.0/-11.0	0.83/0.81
Swarthmore	3577/3576	-3.4/-3.4	0.83/0.80
South Pole	11210/11209	-15.0/-15.0	0.82/0.80
Thule	4521/4521	-4.3/-4.3	0.75/0.73

Figure 7 The hysteresis curves of odd cycles (21 and 23) and corresponding linear fits. Orange and green diamonds show extended (see the text) and interpolated data, respectively.



of these overlapping periods. This method ensures continuity of the model and calculated data.

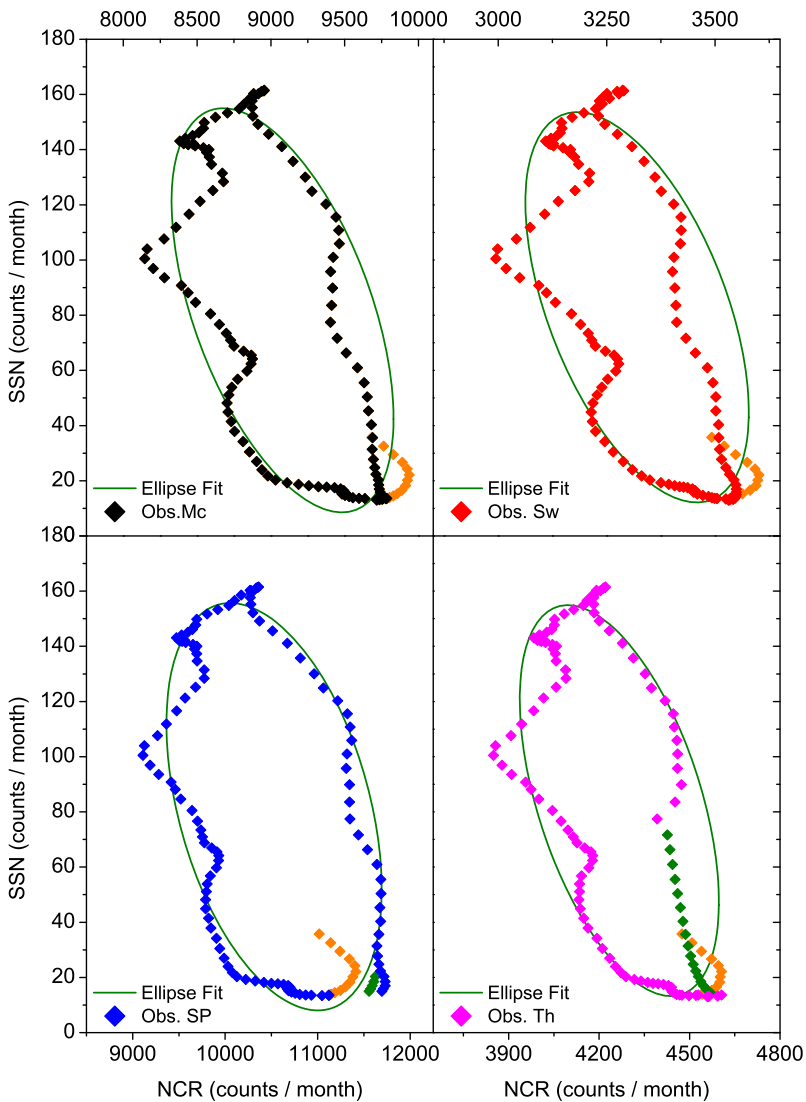


Figure 8 Ellipse fitting of NCRs recorded by four neutron monitors for the 21st cycle. Orange and green diamonds show extended (see the text) and interpolated data, respectively.

4.1. Even Cycles

In order to calculate the NCRs based on SSNs, a linear regression analysis was applied to the data associated with even cycles 20 and 22, because these hysteresis loops are sufficiently narrow to provide a good fit (Figure 6).

The linear relations are shown in Table 3 together with correlation coefficients between observed and calculated NCRs. These correlation coefficients are generally high: 0.91, 0.90, 0.58, and 0.88 for cycle 21 and 0.97, 0.96, 0.97, and 0.96 for cycle 23 for the McMurdo, Swarthmore, South Pole, and Thule NMs, respectively, which imply that the linear model is

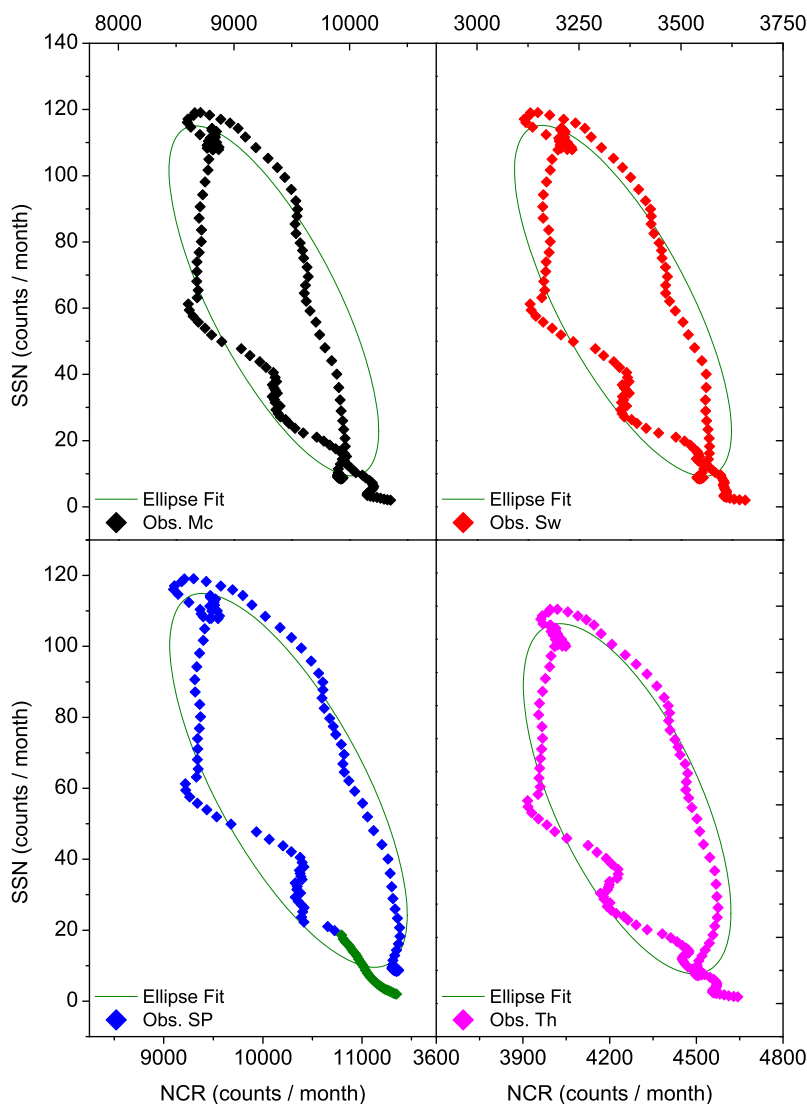


Figure 9 Ellipse fitting of NCRs recorded by four neutron monitors for the 23rd cycle. Green diamonds show interpolated data.

a good representation of the relationship between NCRs and SSNs during even solar cycles. The low correlation coefficient calculated for the South Pole NM results from a gap in the data.

4.2. Odd Cycles

Linear regression analysis was used as a first approach to calculate NCRs during sunspot cycles 21 and 23. The applied linear regression fits (Table 4) are shown in Figure 7 together with the relevant hysteresis loops. It is obvious that linear fits do not provide a good repre-

Table 5 The parameters and correlation coefficients of ellipse modeling of NCRs for odd cycles [Equations (1) and (2)]. The first value in each row shows the result for smoothed data, whereas the second value represents result for non-smoothed data.

Station	Z_1	Z_2	A	B	α	Correlation coefficient
	smoothed / unsmoothed					
Cycle 21						
McMurdo	9079/9082	82/83	753/869	62/63	3.00/2.12	0.97/0.93
Swarthmore	3320/3325	83/84	261/308	59/62	8.81/6.09	0.97/0.92
South Pole	10529/10525	82/82	1164/1282	67/68	1.47/1.17	0.92/0.90
Thule	4267/4270	84/84	332/385	60/62	6.55/4.86	0.97/0.94
Cycle 23						
McMurdo	9345/9364	62/62	908/1052	35/37	2.49/1.94	0.96/0.94
Swarthmore	3358/3354	62/62	269/311	35/39	8.58/6.56	0.95/0.93
South Pole	10258/10229	62/62	1199/1341	37/40	1.82/1.46	0.95/0.93
Thule	4261/4256	62/62	361/407	40/42	5.51/4.43	0.97/0.95

sensation of the relationship between SSNs and NCRs during odd solar cycles, in particular during solar cycle 21 where the correlation coefficients are much lower compared to those associated with even cycles.

The reason for these low correlations associated with the odd cycles is related to the shape of the hysteresis loops, which are significantly wider compared to the even-cycle loops. Therefore, to improve the representation of the relationship between SSNs and NCRs, ellipse modeling was applied to the data associated with the odd cycles (Figures 8 and 9).

The equations of the fitted ellipses in the parametric form are as follows:

$$\begin{bmatrix} \text{NCR} \\ \text{SSN} \end{bmatrix} = \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} + Q(\alpha) \begin{bmatrix} A \cos \theta \\ B \sin \theta \end{bmatrix}, \quad (1)$$

$$Q(\alpha) = \begin{bmatrix} -\cos \alpha & \sin \alpha \\ -\sin \alpha & -\cos \alpha \end{bmatrix}, \quad (2)$$

where Z_1 and Z_2 are centroid points of the fitted tilted ellipse, $Q(\alpha)$ is the rotation matrix, α is the tilt angle, A and B are semi-axes, and $0 \leq \theta \leq 2\pi$. The rotation angle θ is calculated for each data point in the time domain, extending from the beginning of an odd cycle to the first 12 points of the forthcoming cycle, which in turn provides the phase variation in time. The resulting parameters of the ellipse fitting are shown in Table 5 together with the correlation coefficients.

The obtained correlation coefficients between the observed and the calculated values are higher than those of the linear fits applied to odd cycles and as high as the linear approach applied to the even cycles. This justifies the use of ellipse fitting for odd cycles where the hysteresis loops are wider (Tables 4 and 5) and imply that we can obtain good agreement between the observed and the calculated NCRs by using ellipse modeling.

By using different models for individual odd and even cycles, characterized by different hysteresis loop shapes, we calculated the temporal NCR variations observed at four different neutron monitors for cycles 20, 21, 22, and 23. The observed and calculated values are shown in Figure 10 together with corresponding correlation coefficients between NCR_{Obs} and NCR_{Cal} . Using this new model, we obtain the following correlation coefficients over the last four solar cycles: 0.95, 0.95, 0.94, and 0.95 for 7-month moving average smoothed data

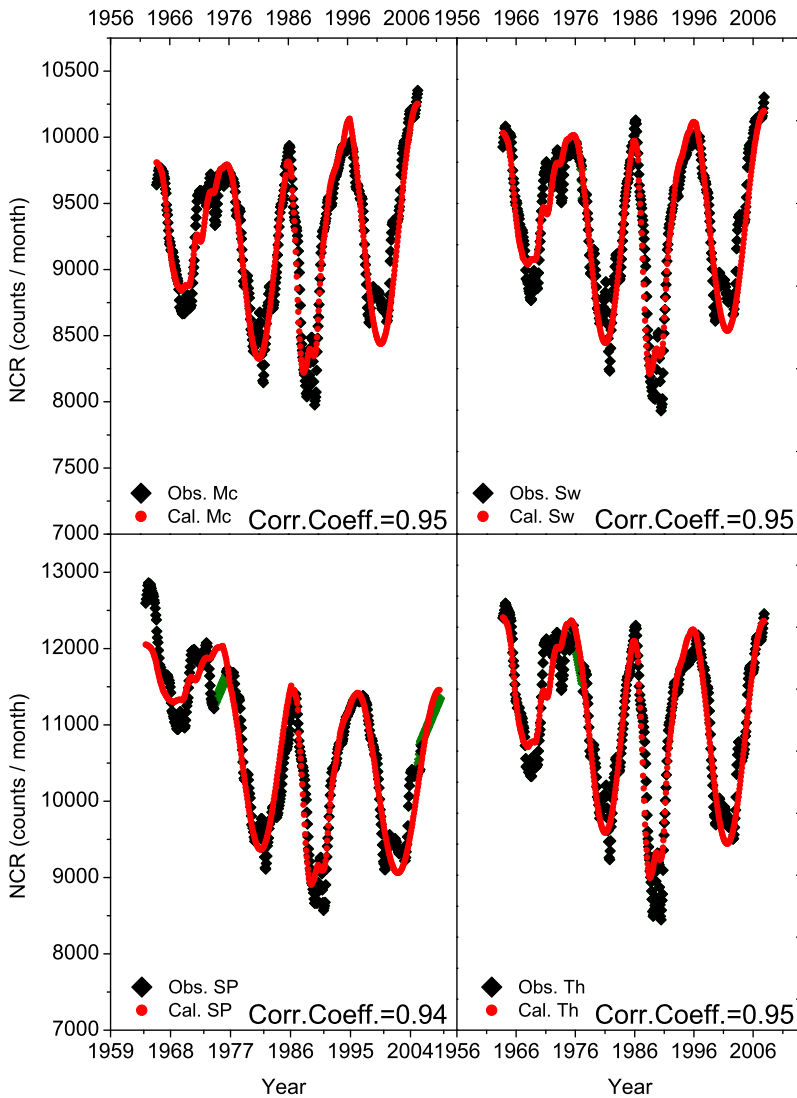


Figure 10 Observed and modeled NCRs variations in time with the relevant correlation coefficients for the neutron monitors. Black diamonds and red dots show observed and calculated values, respectively, whereas green diamonds show interpolated data.

and 0.92, 0.91, 0.92, and 0.93 for non-smoothed data for the McMurdo, Swarthmore, South Pole, and Thule NMs, respectively. Application of linear regression analysis to the full data sets, *i.e.* both even and odd cycles, from the same NMs results in correlation coefficients of 0.84, 0.84, 0.62, and 0.78 for smoothed data and 0.82, 0.82, 0.61, and 0.76 for non-smoothed data, respectively. Comparing the correlation coefficients calculated using ellipse and linear models for the non-smoothed data for odd cycles (*i.e.* the last columns of Tables 4 and 5), we find that ellipse modeling, in general, improves the correlation with 31 %. In

other words, it is clear that inclusion of the hysteresis effect thus provides a substantially improved representation of the relationship between NCRs and SSNs.

5. Discussion and Conclusions

In the present work, we suggest a new empirical approach to model the link between NCRs and SSNs based on the hysteresis effect. Instead of using linear regression analysis, which is generally applied to both odd and even cycles in order to reconstruct NCRs and GCR intensity (see the multi-component analysis of Mavromichalaki and Petropoulos (1987) and Mavromichalaki, Marmatsouri, and Vassilaki (1990)), we used linear- and ellipse-modeling procedures separately for even and odd cycles, respectively. This approach relies on the fact that the hysteresis curves for even and odd solar cycles exhibit different loop shapes, which is related to the polarity states of the Sun. During even cycles, the hysteresis loops are narrow which favors linear modeling. However, the loops are relatively wide during odd cycles, where ellipse modeling provides a better representation of the data. By using this method, we reconstructed NCRs based on SSNs during solar cycles 20, 21, 22, and 23 (Figure 10). The NCRs obtained using the new approach are in good agreement with observed NCRs, as indicated by correlation coefficients of 0.95, 0.95, 0.94, and 0.95 for smoothed data from the neutron monitors McMurdo, Swarthmore, South Pole, and Thule, respectively. In comparison, we obtained correlation coefficients of 0.84, 0.84, 0.62, and 0.78 for smoothed data from the same NMs when applying linear fit to the whole data sets.

We emphasize that this study focuses on modeling the relationship between secondary cosmic ray particles, primarily neutrons, as an indicator of GCR intensity, and SSNs using a parameter that can be reconstructed for earlier periods. A major goal of this ongoing research is thus to improve our understanding of the relationship between high-resolution cosmogenic radionuclide (^{10}Be and ^{14}C) data, NCRs, and SSNs, in order to extend our knowledge of solar activity back in time beyond the interval from which instrumental data are available.

Acknowledgements Funding for the Stellar Astrophysics Centre is provided by the Danish National Research Foundation (Grant agreement no.: DNR106). MFK, CK and JO acknowledged support from the Carlsberg Foundation. Neutron monitors of the Bartol Research Institute are supported by the United States National Science Foundation (Grant agreement numbers: ANT-0739620 and ANT-0838839), and by the University of Delaware Department of Physics and Astronomy and Bartol Research Institute.

References

- Ahluwalia, H.S., Ygbuhay, R.C.: 2012, *Adv. Space Res.* **49**, 493.
- Alanko-Huotari, K., Usoskin, I.G., Mursula, K., Kovaltsov, G.A.: 2007, *J. Geophys. Res.* **112**, 8101.
- Bard, E., Frank, M.: 2006, *Earth Planet. Sci. Lett.* **248**, 1.
- Baroni, M., Bard, E., Petit, J.-R., Magand, O., Bourlès, D.: 2011, *Geochim. Cosmochim. Acta* **75**, 7132.
- Belov, A.: 2000, *Space Sci. Rev.* **93**, 79.
- Chowdhury, P., Kudela, K., Dwivedi, B.N.: 2013, *Solar Phys.* **286**, 577.
- Dunai, T.J.: 2010, *Cosmogenic Nuclides: Principles, Concepts and Applications in the Earth Surface Sciences*, Cambridge University Press, Cambridge, 2.
- Gupta, M., Mishra, V.K., Mishra, A.P.: 2006, *J. Astrophys. Astron.* **27**, 455.
- Hale, G.E., Ellerman, F., Nicholson, S.B., Joy, A.H.: 1919, *Astrophys. J.* **49**, 153.
- Hathaway, D.H.: 2010, *Living Rev. Solar Phys.* **7**(1). <http://solarphysics.livingreviews.org/Articles/lrsp-2010-1/>.
- Heikkilä, U., Beer, J., Jouzel, J., Feichter, J., Kubik, P.: 2008, *Geophys. Res. Lett.* **35**, 5817.
- Hoyt, D.V., Schatten, K.H.: 1993, *J. Geophys. Res.* **98**, 18895.
- Jokipii, J.R.: 2011, In: *The 2011 Heliophysics Summer School*, Boulder. http://www.vsp.ucar.edu/Heliophysics/pdf/2011_Jokipii_Cosmic-Rays-Lecture.pdf.

- Jokipii, J.R., Levy, E.H., Hubbard, W.B.: 1977, *Astrophys. J.* **213**, 861.
- Lean, J.: 2000, *Geophys. Res. Lett.* **27**, 2425.
- Mavromichalaki, H., Petropoulos, B.: 1984, *Astrophys. Space Sci.* **106**, 61.
- Mavromichalaki, H., Petropoulos, B.: 1987, *Earth Moon Planets* **37**, 79.
- Mavromichalaki, H., Belehaki, A., Rafios, X., Tsagouri, I.: 1996, *Astrophys. Space Sci.* **246**, 7.
- Mavromichalaki, H., Marmatsouri, E., Vassilaki, A.: 1990, *Solar Phys.* **125**, 409.
- Mavromichalaki, H., Paouris, E., Karalidi, T.: 2007, *Solar Phys.* **245**, 369.
- Marmatsouri, L., Vassilaki, A., Mavromichalaki, H., Petropoulos, B.: 1995, *Adv. Space Res.* **16**, 245.
- Mavromichalaki, H., Belehaki, A., Rafios, X.: 1998, *Astron. Astrophys.* **330**, 764.
- Nagashima, K., Morishita, I.: 1980, *Planet. Space Sci.* **28**, 195.
- Otaola, J.A., Perez-Enriquez, R., Valdes-Galicia, J.F.: 1985, *19th Int. Cosmic Ray Conf.* **4**, 493.
- Owens, M.J., Usoskin, I., Lockwood, M.: 2012, *Geophys. Res. Lett.* **39**, 19102.
- Paouris, E., Mavromichalaki, H., Belov, A., Gushchina, R., Yanke, V.: 2012, *Solar Phys.* **280**, 255.
- Parker, E.N.: 1965, *Planet. Space Sci.* **13**, 9.
- Potgieter, M.S., Burger, R.A., Ferreira, S.E.S.: 2001, *Space Sci. Rev.* **97**, 295.
- Singh, M., Singh, Y.P., Badruddin: 2008, *J. Atmos. Solar-Terr. Phys.* **70**, 169.
- Solanki, S.K., Usoskin, I.G., Kromer, B., Schüssler, M., Beer, J.: 2004, *Nature* **431**, 1084.
- Usoskin, I.G., Kananen, H., Mursula, K., Tanskanen, P., Kovaltsov, G.A.: 1998, *J. Geophys. Res.* **103**, 9567.
- Usoskin, I.G., Gladysheva, O.G., Bobik, P., Kudela, K., Kananen, H.: 1999, *Czechoslov. J. Phys.* **12**, 49.
- Usoskin, I.G., Bobik, P., Gladysheva, O.G., Kananen, H., Kovaltsov, G.A., Kudela, K.: 2001, *Adv. Space Res.* **27**, 565.
- Van Allen, J.A.: 2000, *Geophys. Res. Lett.* **27**, 2453.
- Webber, W.R., Lockwood, J.A.: 1988, *Geophys. Res. Lett.* **93**, 8735.